LDMS USER GROUP MEETING

August 24, 2026

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Agenda

- V4.5.3 Release
- V4.6.1 Status Updates
- Q&A

V4.5.3 Release

V4.5.3 Is Released

- Tagged and released last week, available now on Github https://github.com/ovis-hpc/ldms/releases/tag/v4.5.3

V4.5.3: What We Already Previewed

- Networking:
  - Improved address-family selection for reliable connections across IPv4-only, IPv6-only, and mixed environments
- Python/Cython:
  - Added Cython 3.12 build compatibility
  - Improved memory cleanup for LDMS Set objects
- slurm_notifier Enhancement:
  - Added WORKFLOW_ID support
  - Added a `message_tag` option
- Test framework:
  - Aligned message tag terminology across examples (`channel` -> `tag`)
- Consolidated validation checks in the `ovis_map` collection library
- Corrected type declaration in the rail IB connection message struct

V4.5.3: Added Since Our Last Update

- `linux_proc_sampler`:
  - New `/proc/pid/maps` data, per-pid-set control, quieter defaults, plus new test coverage (with and without SOS storage)
- Build hardening:
  - Resolved GCC -Werror/gnu23 strictness issues that surfaced on newer toolchains (openSUSE 16, Ubuntu)
- Documentation:
  - Updated `app_sampler`/`linux_proc_sampler` man pages to match the new options

Post Release & V4.5.4

- Issues surfaced from build/environment reports
  - Python/Cython build-warning issues, not runtime or data-correctness bugs
    - Cython enum-export warning: Cython 3.0.x raised a UserWarning because `ldms_xprt_event_type` was declared `cdef` while `ldms.pyx` was also trying to export it as a Python-visible enum class
    - Cython performance hint: a separate build-time hint from Cython's optimizer was showing up as noise during builds; suppressed so it doesn't distract from real warnings
  - Issues already addressed on b4.5

Post Release & V4.5.4 (cont.)

- John (NERSC) observed memory growth of a container that runs a Python script that subscribes to LDMS Message via LDMS Python API and forwards them to Kafka
  - Under investigation
  - Conjecture:
    - The LDMS Python API delivers received messages on an IO thread, which posts them to an internal queue; the script's main thread drains that queue and calls kafka.send(). If kafka.send() blocks or falls behind, the main thread can't drain the queue as fast as the IO thread fills it, so the queue — and memory usage — grows over time

V4.6.1 Updates

V4.6.1: Status

- Currently in testing – accepting only bug fixes at this point
- Features and functionality in testing
  - LDMS Message
  - Event-based Storage
  - Slurm-related regression testing

Q & A

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Thank you!